

Portfolio Platform

Full-stack regulated enterprise template

NestJS Clean Architecture API + Next.js FSD admin & marketing site

Problem

- Portfolio / marketing sites need **CMS, RBAC admin, and compliance controls**
- Teams rebuild auth, audit, MFA, and public API boundaries on every project
- CTOs need to assess architecture **in minutes**, not days of code reading

Solution

Two paired open-source templates:

Repo	Role
<code>nestjs-fsd-portfolio-template</code>	CMS API, auth, RBAC, audit
<code>nextjs-fsd-portfolio-template</code>	Marketing site + admin + BFF

Fork → configure white-label env → deploy

C1 — System context

```
flowchart LR
    Visitor[Public visitor] --> Web[Next.js App]
    Admin[Admin] --> Web
    Web -->|private| API[NestJS API]
    API --> MySQL[(MySQL)]
    API --> Redis[(Redis)]
    Web --> IdP[OIDC IdP]
```

C2 — Containers

- **Marketing SSR** — public pages, CMS content, localized EN/MN
- **Admin dashboard** — FSD features, permission-gated CRUD
- **BFF** — httpOnly JWT, CSRF, allowlisted proxy
- **NestJS API** — Clean Architecture, 24+ controllers
- **MySQL + Redis** — persistence, token blacklist, permission cache

Security model

- BFF: tokens never in `localStorage`
- RBAC: permission codes on API + UI guards
- MFA: TOTP enrollment + step-up for sensitive roles
- CSRF on mutating BFF routes
- Audit interceptor on admin mutations
- GDPR export & erasure workflows

CMS capabilities

- Projects, skills, experiences
- Blog, pricing, navigation tree
- Site settings (hero, SEO, logos, contact)
- Localized content (EN/MN)
- Media upload (S3-compatible)

White-label in 5 minutes

Env (deploy-time): `APP_DISPLAY_NAME`, `MFA_ISSUER`, `NEXT_PUBLIC_APP_NAME`

CMS (runtime): logos, site name, SEO, hero

i18n: `{brandName}` placeholders in marketing copy

See `docs/WHITE-LABEL.md`

Enterprise controls

29-item regulated enterprise checklist with E2E evidence:

- Session refresh + expiry dialog
- BFF allowlist tests
- MFA + OAuth smoke tests
- Audit log retention
- Helm lint in CI

Deployment options

Target	Pattern
Railway	2 services + MySQL + Redis plugins
Kubernetes	Helm chart <code>portfolio-stack</code>
Vercel	Frontend + hosted API

Same env contract across targets

Observability

- OpenTelemetry (opt-in) — API + frontend
- Sentry DSN (opt-in)
- Structured logging (Winston)
- Health + readiness probes

Adoption path

1. Fork both repos
2. Set brand env vars + run seed
3. Upload logos in site settings
4. Deploy API + frontend
5. Enable MFA / OIDC for production

Links

- API: github.com/devTugu/nestjs-fsd-portfolio-template
- Web: github.com/devTugu/nextjs-fsd-portfolio-template
- C4 docs: `docs/SYSTEM-ARCHITECTURE.md`
- ADRs: `docs/adr/`
- Runbook: `docs/RUNBOOK.md`

Generate PDF: `npm run docs:pitch-pdf`